
AWS Prescriptive Guidance

Migrating on-premises Teradata environments to Teradata VantageCloud Lake on AWS



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Migrating on-premises Teradata environments to Teradata VantageCloud Lake on AWS

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Overview

This guide provides an overview of options and considerations for migrating your Teradata Vantage on-premises system to Teradata VantageCloud Lake on AWS. [Teradata VantageCloud Lake on AWS](#) combines the advanced analytics platform of Teradata Vantage with the global reach and scalability of AWS infrastructure. You can use this guide to gain a better understanding of the data migration process and how to choose a data migration approach that meets your organization's business and IT requirements. The guide also includes reference architectures that you can use as templates to build your own migration architecture.

This guide is intended for technical decision-makers, business decision-makers, and [AWS Services Partners](#).

Targeted business outcomes

This guide can help you and your organization achieve the following objectives:

- Understand the costs, licensing requirements, and prerequisites for migrating to Teradata VantageCloud Lake in the AWS Cloud.
- Identify a reference architecture that meets your organization's requirements.
- Identify a migration approach that meets your organization's requirements.

Costs and licenses

Every organization has its own unique data analytics requirements. Teradata VantageCloud Lake offers a choice of [cloud pricing models](#) so that you can find the right fit for your workloads, budget, and goals.

Contact your Teradata account team for more information about licensing.

Prerequisites

Before you migrate from Teradata Vantage to the AWS Cloud, we recommend that you meet the following prerequisites:

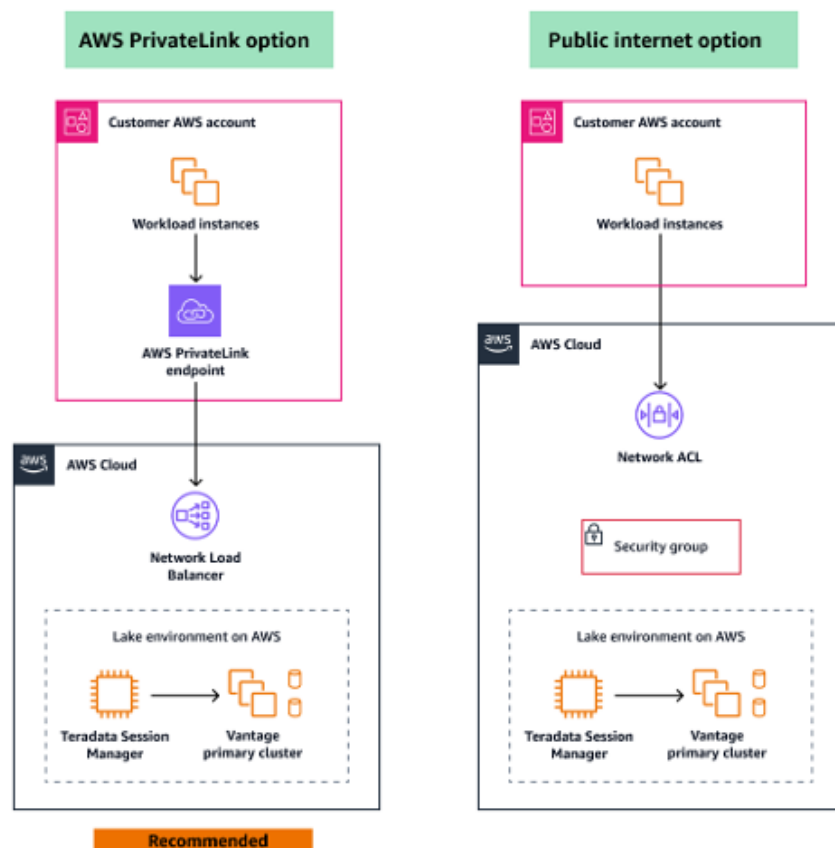
- Update your on-premises ecosystem where possible, including upgrading to the latest versions of Teradata Tools and Utilities (TTU) and third-party applications.
- Identify resources required for the migration process. The [Migration process](#) section of this guide covers resource requirements in more detail.

Connectivity options

There are two supported connectivity options for connecting to Teradata VantageCloud Lake:

- AWS PrivateLink (recommended) – connect from cloud to cloud. You can use this option to connect your AWS account to VantageCloud Lake.
- Public internet – connect over the internet. You can use this option to connect your on-premises system to VantageCloud Lake.

The following diagram shows these supported connectivity methods.

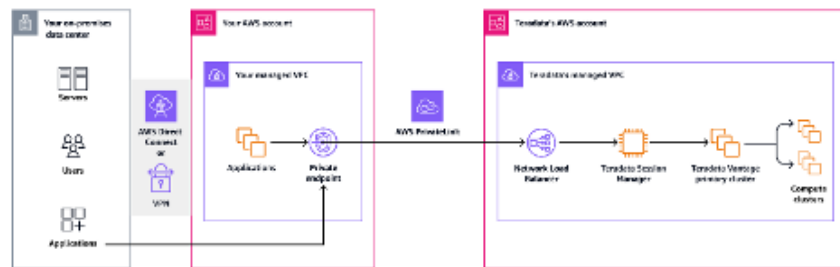


PrivateLink architecture

AWS PrivateLink provides connectivity between virtual private clouds (VPCs). You can access Teradata VantageCloud Lake over private IP addresses from your virtual network while keeping the data flow on the AWS secure backbone network. Data never traverses the public internet. This significantly reduces exposure to common security threats.

PrivateLink allows only unidirectional network connectivity. Applications that require a connection to be initiated from both endpoints require two PrivateLink connections.

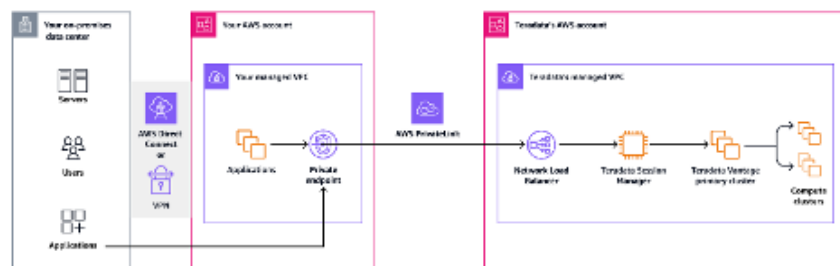
The following diagram shows a PrivateLink architecture where a private endpoint in an AWS account uses PrivateLink to connect to Teradata Session Manager, which connects to a Vantage primary cluster.



For more information, see [AWS PrivateLink](#) or contact your [Teradata account team](#).

Public internet

If your architecture requires hybrid connectivity from on premises to Teradata VantageCloud Lake, you can use the public internet connectivity option. You can also use this option to connect from another VPC over the internet. You control the allowed CIDR ranges. The following diagram shows a customer's VPC using the internet to connect to Teradata-managed VantageCloud Lake.



Migration process

An enterprise that implements a migration strategy doesn't typically transfer all of its data to the cloud at one time. In fact, many organizations take an iterative approach to cloud migration, driven by both business and technical demands. For more information about deployment options, see the article [On-premises to Cloud Migration](#) on the Teradata website.

When you are ready to migrate your data to VantageCloud Lake, the Teradata migration framework will guide you through each step. The framework covers the entire span of a cloud migration—from design through go-live, from the Teradata platform to the ecosystem connecting to it—and provides guidance for internal and external teams. It is a reliable, fast, and effective roadmap for migrating to, and taking advantage of, Teradata's cloud offerings.

The migration framework consists of the following phases:

- **Design**
- **Connect**
- **Execute**
- **Go live**

Design

The goals of the Design phase are to determine your migration readiness and secure agreement with your stakeholders on a common migration approach. This phase consists of the following stages:

- **Initiate** – set stakeholder expectations.
- **Assess** – complete a high-level assessment.
- **Design solution** – complete a detailed assessment, design, and approach.

We recommend that you engage these roles during this phase to provide input to the requirements:

- Your project sponsor and business users
- Your data analytics architect and ecosystem architect
- Your information security architect
- Your data warehouse lead
- Your test lead

The following Teradata roles are involved in this phase and will lead the requirement discussions:

- Teradata account team
- Teradata solution engineer
- Teradata cloud architect

At the end of the Design phase, you will have detailed architecture diagrams, a high-level timeline, an initial list of critical dependencies and risks, and agreement on the overall migration approach.

Connect

The goals of the Connect phase are to confirm your migration readiness, deploy your cloud instances, and complete detailed planning. This phase consists of the following stages:

- Test strategy and success criteria – define what success looks like.
- Onboarding – provision and deploy your cloud instances.
- Implementation plans – define a detailed implementation plan.

We recommend that you engage these roles during this phase to provide input to the detailed planning:

- Your project sponsor and business users
- Your data analytics architect and ecosystem architect
- Your information security architect

- Your data warehouse lead
- Your IT leads for areas such as firewalls and network connectivity
- Your test lead
- Your project manager

The following Teradata roles are involved in this phase and will lead the requirement discussions:

- Teradata solution engineer
- Teradata cloud architect
- Teradata service experience manager
- Teradata cloud provisioning team
- Teradata cloud support teams
- Teradata project manager
- Teradata migration consultant
- Teradata cloud implementation manager

At the end of the Connect phase, you will have a test strategy, deployed cloud instances, a detailed implementation plan, and an agreement that you are ready to begin implementing your migration plan.

Execute

The goals of the Execute phase are to prepare an environment to support your testing, complete all pre-migration testing, and implement the migration plan. This phase consists of the following stages:

- Convert – migrate test data to the VantageCloud Lake systems and convert extract, transform, and load (ETL) or business intelligence (BI) workloads if required.

- Validate platform and ecosystem – regression test any converted ETL or BI workloads, test Teradata and third-party applications, and complete performance testing.
- Migrate, validate, and catch-up data – implement detailed migration plan.

We recommend that you engage these roles during this phase to complete the migration:

- Your project sponsor and business users
- Your data warehouse lead
- Your ETL and BI leads
- Your test lead and testing teams
- Your IT leads for areas such as firewalls and network connectivity
- Your project manager

The Teradata roles that are involved in this phase include:

- Teradata migration consultant
- Teradata service experience manager
- Teradata cloud support teams
- Teradata project manager
- Teradata migration consultant
- Teradata cloud implementation manager

At the end of the Execute phase, your systems will be live in the AWS VantageCloud Lake.

Go Live

The goal of the Go Live phase is to optimize the cloud system and transition to operational support. This phase consists of the following stages:

- Optimize – complete workload management and other post-migration work.
- Transition – complete transition to operational support teams.

We recommend that you engage these roles during this phase:

- Your data warehouse lead
- Your IT leads for areas such as firewalls and network connectivity
- Your test lead
- Your project manager

The following Teradata roles are involved in this phase:

- Teradata service experience manager
- Teradata cloud support teams
- Teradata project manager
- Teradata cloud implementation manager

At the end of the Go Live phase, your VantageCloud Lake system will be live on AWS, ongoing operational support will begin, and post-migration activities will be complete.

Data migration approaches

This section provides an overview of the most common migration approaches, and highlights some of the key benefits and challenges of each approach. The following tools are used in the data migration approaches covered in this section:

- Teradata Data Transfer Utility (DTU) – offline migration
- Network attached storage (NAS) cloud storage device – physical data migration using AWS Snowball

Use the following table to identify the migration approach that meets your requirements.

Approach	Description	Benefits	Challenges
Teradata DTU with systems offline (recommended)	Uses DTU streaming to migrate data from the on-premises source to the AWS target environment.	• Maximizes throughput and minimizes the impacts of network and database restarts on migration. • Offers the fastest elapsed time for migration. • Streamlines support from the data migration,	Source system outage during migration

		global support, and engineering teams.	
Teradata DSA migration using a NAS device	Uses DSU to migrate data from the source to a NAS cloud storage device (for example, AWS Snowball), transport the cloud storage device to a cloud data center, and offload to cloud storage (for example, to Amazon S3). Or, it uses DSU to migrate data from cloud storage to the target environment.	Workaround for poor network capacity to the cloud	• Logistics, risk of damage to storage device in transport, multiple migrations might be required. • Longer elapsed time. • Requires significant effort to synchronize target with source systems after migration.

Tooling

Teradata offers the following tools:

- [Teradata VantageCloud Lake Console](#) provides an integrated user experience for analysts, developers, and operations.
- [Teradata Data Mover](#) enables you to automate, control, and monitor the data movement between Teradata environments.
- [Teradata QueryGrid](#) enables analytics across a heterogeneous analytical ecosystem.
- [Teradata Tools and Utilities \(TTU\)](#) provides a full collection of Teradata client tools, including load and unload utilities, open interfaces, and connectivity drivers.

Next steps

We recommend that you consider the following next steps:

- [Learn how organizations use Teradata Vantage](#) to connect and analyze data across their entire ecosystem, including data lakes, object stores, and cloud services.
- Identify the migration workloads in your on-premises environments or other cloud environments. Then, review your [deployment options](#) and identify your destination environment. For more information, see [Teradata Vantage on AWS](#) in the Teradata documentation.
- Choose either a rehost (that is, lift and shift) or replatform (that is, make some workload changes) migration strategy. For more information, see [Lift-and-shift Cloud Migration](#) and [A Guide to Cloud Migration](#) in the Teradata documentation.
- Assess the costs of running Teradata VantageCloud on AWS. For more information, see [Teradata VantageCloud Pricing](#) in the Teradata documentation.
- Make sure that you meet all the prerequisites for a migration based on the [Prerequisites](#) section of this guide.

Note: Teradata makes a commercially reasonable effort to make Teradata Vantage available on AWS systems all day, every day. Teradata targets an SLA (excluding planned downtime) of 99.9 percent measured monthly. For operating model and services options, see [Teradata Vantage on AWS](#) in the Teradata documentation.

Resources

For more information, see the following resources on the Teradata website.

- [Teradata and partners](#)
- [Unlock data, activate analytics, accelerate value](#)
- [Teradata VantageCloud Lake pricing](#)
- [Teradata VantageCloud Lake deployment options](#)
- [Teradata VantageCloud Lake and ClearScape Analytics: Empowering AI Innovation](#)
- [Risk mitigation](#)
- [Operational excellence](#)
- [Asset optimization](#)